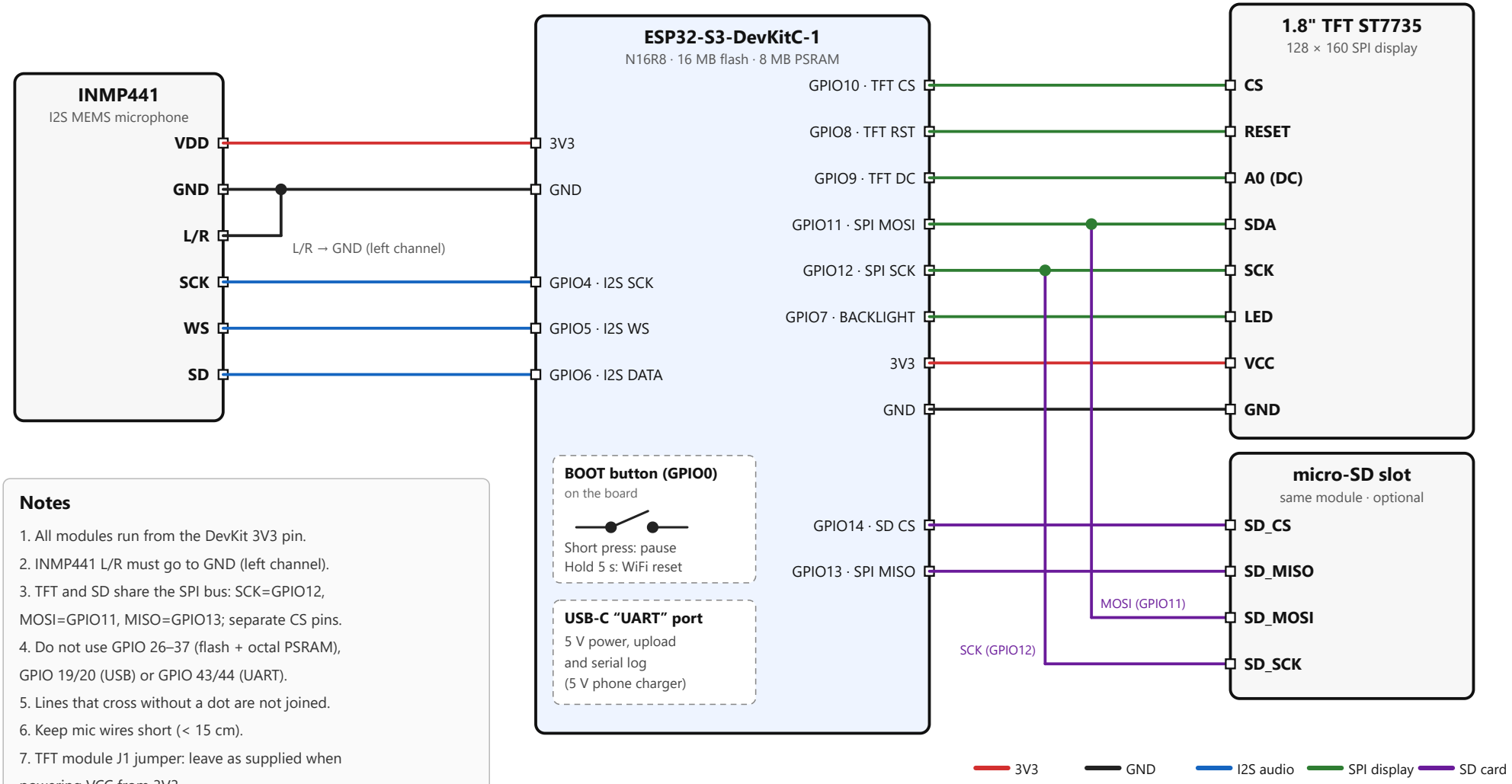
Naam Jap Counter — Schematic (ESP32-S3-DevKitC-1)

ESP32-S3-DevKitC-1 N16R8 · INMP441 I2S microphone · 1.8" ST7735 TFT with micro-SD · build target esp32-s3-n16r8

Rev 2.1

Date 11-11-2026 · Sheet 1 of 2



Notes

1. All modules run from the DevKit 3V3 pin.
2. INMP441 L/R must go to GND (left channel).
3. TFT and SD share the SPI bus: SCK=GPIO12, MOSI=GPIO11, MISO=GPIO13; separate CS pins.
4. Do not use GPIO 26–37 (flash + octal PSRAM), GPIO 19/20 (USB) or GPIO 43/44 (UART).
5. Lines that cross without a dot are not joined.
6. Keep mic wires short (< 15 cm).
7. TFT module J1 jumper: leave as supplied when powering VCC from 3V3.

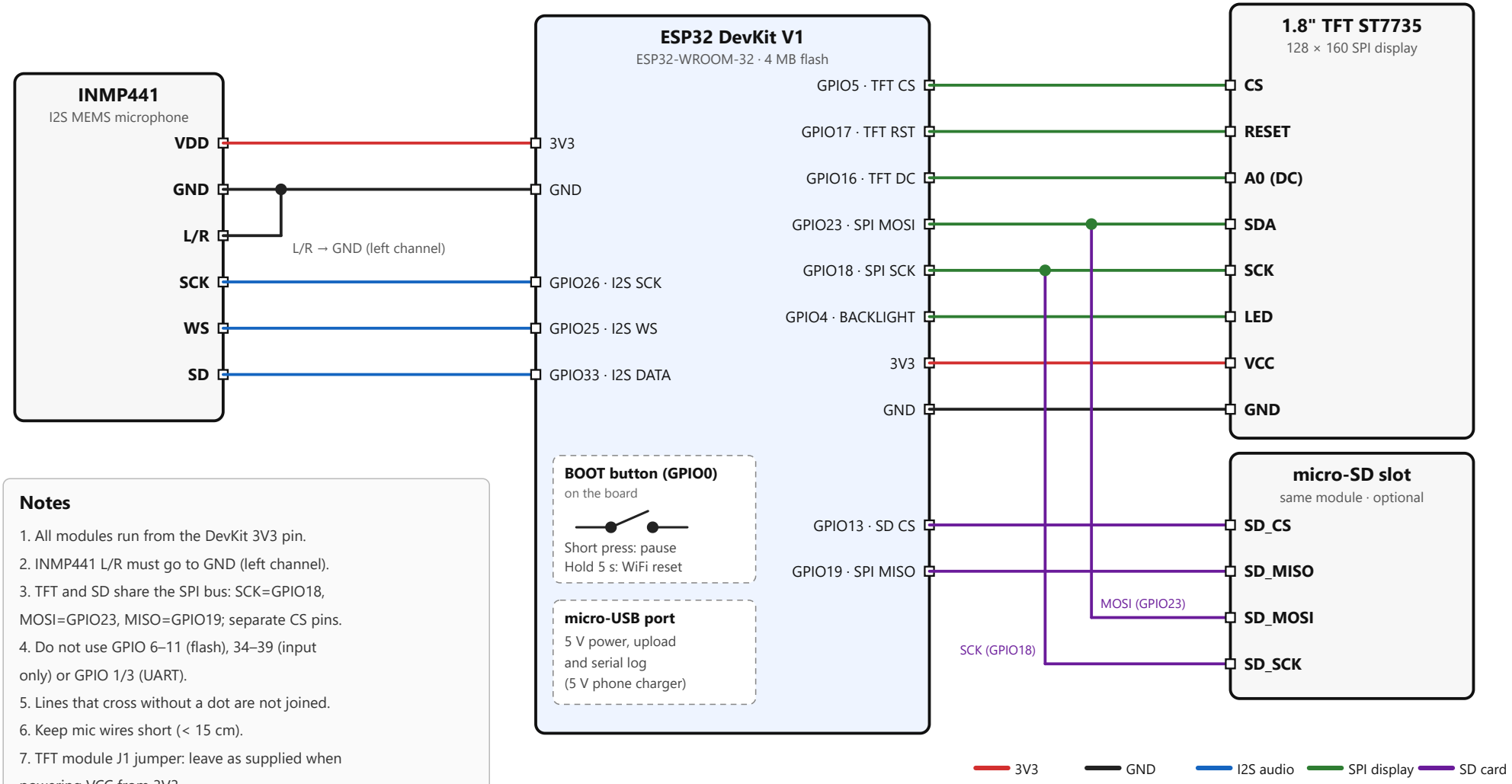
Pin map lives in include/config.h

Naam Jap Counter — Schematic (ESP32 DevKit V1)

ESP32 DevKit V1 ESP32-WROOM-32 · INMP441 I2S microphone · 1.8" ST7735 TFT with micro-SD · build target esp32dev

Rev 2.1

Date 11-11-2026 · Sheet 2 of 2



Notes

1. All modules run from the DevKit 3V3 pin.
2. INMP441 L/R must go to GND (left channel).
3. TFT and SD share the SPI bus: SCK=GPIO18, MOSI=GPIO23, MISO=GPIO19; separate CS pins.
4. Do not use GPIO 6–11 (flash), 34–39 (input only) or GPIO 1/3 (UART).
5. Lines that cross without a dot are not joined.
6. Keep mic wires short (< 15 cm).
7. TFT module J1 jumper: leave as supplied when powering VCC from 3V3.

Pin map lives in include/config.h

Wiring table

Every connection on the schematics, one wire per row. Use the column for your board, and choose the matching build target in PlatformIO (`esp32dev` or `esp32-s3-n16r8`). Use short jumper wires (under 15 cm), especially for the microphone.

Module	Module pin	ESP32 DevKit V1	ESP32-S3-DevKitC-1	Purpose
INMP441	VDD	**3V3**	**3V3**	Power 3.3 V
INMP441	GND	**GND**	**GND**	
INMP441	L/R	**GND**	**GND**	Selects left channel
INMP441	SCK	**GPIO 26**	**GPIO 4**	I2S bit clock
INMP441	WS	**GPIO 25**	**GPIO 5**	I2S word select
INMP441	SD	**GPIO 33**	**GPIO 6**	I2S data (mic → ESP)
TFT	VCC	**3V3**	**3V3**	Power
TFT	GND	**GND**	**GND**	
TFT	CS	**GPIO 5**	**GPIO 10**	Display chip select
TFT	RESET	**GPIO 17**	**GPIO 8**	Display reset
TFT	A0	**GPIO 16**	**GPIO 9**	Data / command
TFT	SDA	**GPIO 23**	**GPIO 11**	SPI MOSI (shared)
TFT	SCK	**GPIO 18**	**GPIO 12**	SPI clock (shared)
TFT	LED	**GPIO 4**	**GPIO 7**	Backlight, PWM brightness
SD slot	SD_CS	**GPIO 13**	**GPIO 14**	SD chip select
SD slot	SD_MOSI	**GPIO 23**	**GPIO 11**	Same wire as TFT SDA
SD slot	SD_MISO	**GPIO 19**	**GPIO 13**	SPI MISO
SD slot	SD_SCK	**GPIO 18**	**GPIO 12**	Same wire as TFT SCK
Board	BOOT button	**GPIO 0**	**GPIO 0**	On board, nothing to wire

> **ESP32-S3-DevKitC-1 pins to avoid:** GPIO 26–37 (used by the 16 MB flash and 8 MB octal PSRAM), GPIO 19/20 (native USB), GPIO 43/44 (UART0 serial log), and GPIO 3, 45, 46 (boot strapping).

> **ESP32 DevKit V1 pins to avoid:** GPIO 6–11 (used by the flash chip), GPIO 34–39 (input only), GPIO 1/3 (UART0 serial log), and GPIO 2, 12, 15 (boot strapping).